



14-848 Cloud Infrastructure

Lecture 2: Cloud Service Models and Virtualization

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Agenda

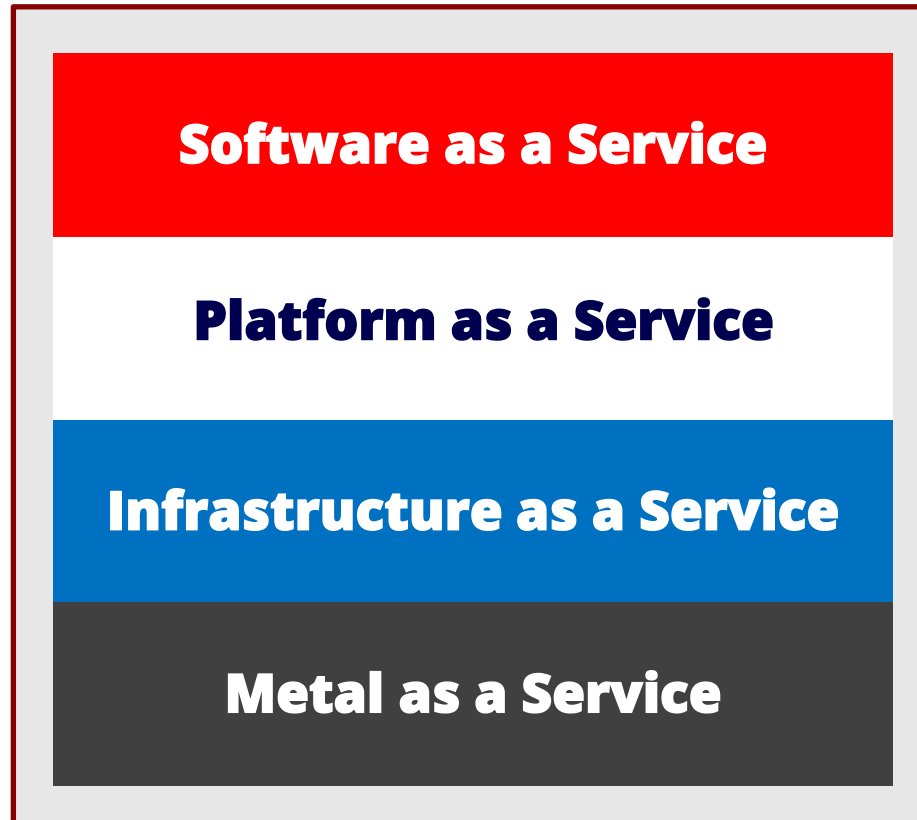
- Announcements
- Cloud Service Models
- Virtualization



Announcements

- HW-1 is released after today's lecture (8:30pm ET / 5:30pm PT).
 - Read its instructions carefully.
 - Pay attention to the unique identifications required to be included in every screenshot.
 - Read the permitted AI usage section carefully.
 - HW-1 is due next Thursday 9/3.

Cloud Service Models



SaaS: Software as a Service

- Provided with access to application software in the cloud
 - On-demand software
- Most applications can be run directly from web browser
- Largest cloud market
- **Examples:** Google Apps, Microsoft Office 365, Oracle's Netsuite, SAP's Concur, Cisco WebEx, GoToMeeting



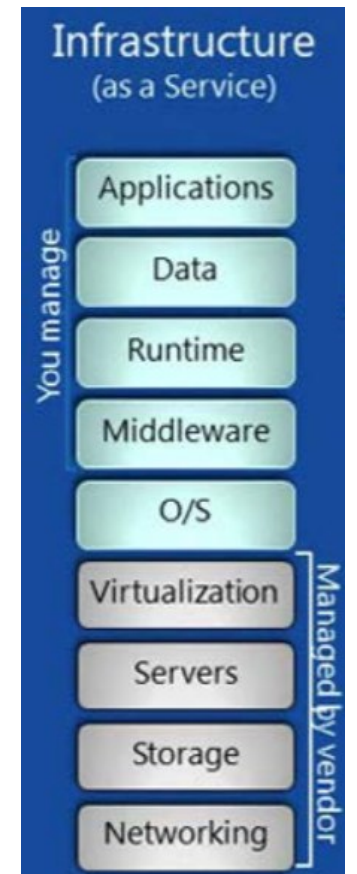
PaaS: Platform as a Service

- Provides computing platforms which typically includes operating system, programming language, execution environment, database, web server, etc. to build cloud applications.
- Applications using PaaS inherit cloud characteristic such as scalability, high-availability, multi-tenancy, SaaS enablement, and more.
- **Examples:** Google App Engine, AWS Elastic Beanstalk, Salesforce.com, Amazon EMR, MS Azure HDInsight, GCP Dataproc.



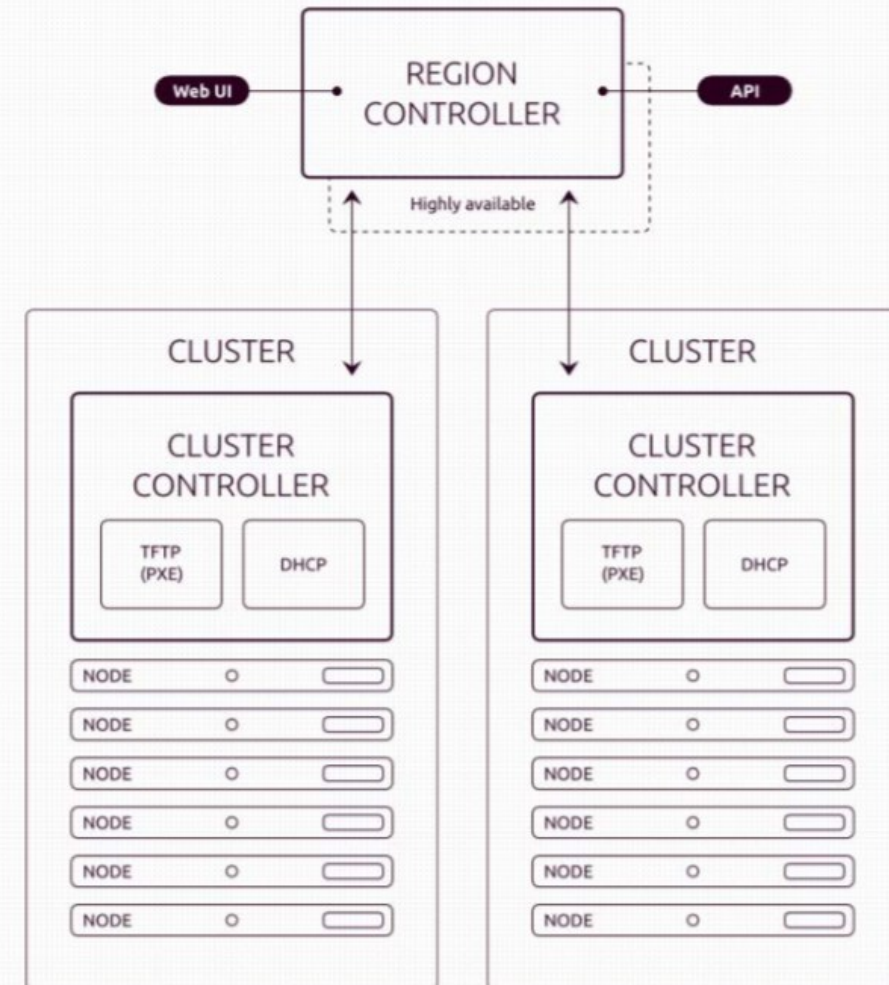
IaaS: Infrastructure as a Service

- Offers storage and computing resources that developers and IT organization use to deliver custom business solutions
- **Examples:** Amazon EC2, VMWare vCloud, GCP Compute Engine

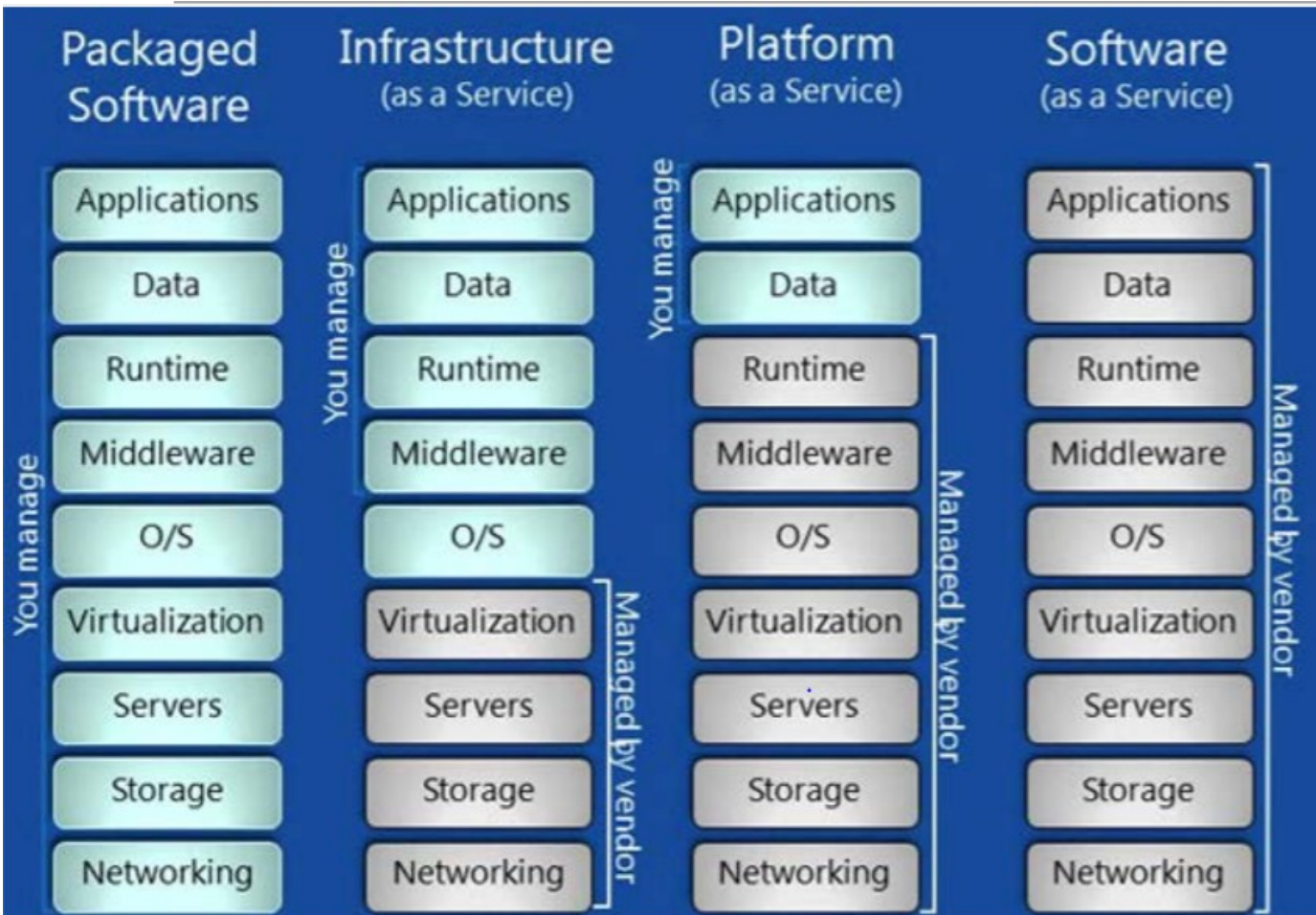


MaaS: Metal as a Service

- Combines the flexibility and scalability of the cloud with the ability to harness the power of physical servers.
- **Example:** Juju
- For more information, watch this video (optional):
https://www.youtube.com/watch?time_continue=280&v=FBCKCO45xlw



Cloud Computing Service Models



In MAAS

- You will have the option to control everything!



PaaS or IaaS?

Vendor Lock-in: the ability to use “what you manage” in cloud environment with different cloud provider.

- PaaS may lock-in applications by requiring users to develop apps based on their specific APIs.
- If you are using PaaS, it might be difficult to switch to different vendor.

Development Tools

- PaaS providers usually allow a set of development tools for their users to shorten development time.
- Another trick for vendor lock-in!



Virtualization



Why to study Virtualization?

Cloud Infrastructure = Data Center + Virtualization

- In this lecture, we will look at Virtualization at a high-level



What is Virtualization?

- Virtualization abstracts the hardware of computing infrastructure into several different execution environments.
 - It creates the illusion that each separate environment is running on its own private computing infrastructure
 - It makes servers, workstations, storage, network and other systems independent of the physical hardware layer
- Virtualization is the fundamental technology that powers Cloud Infrastructure!
 - Virtual resources can be started and stopped easily and quickly.



Virtualization - Definitions

Virtualization

- The process of creating a virtual version of a physical object.

Virtual Machine

- Visual representation of a physical machine (Not JVM).

Virtual Machine Monitor (VMM) or Hypervisor

- A process that separates a computer's operating system and applications from the underlying physical hardware.
- Hypervisor monitors and manages running virtual machines.

Host Machine

- The physical machine that a virtual machine is running on.

Guest Machine

- The virtual machine, running on the host machine.



Why Virtualization is Important?

The following video answers this question:

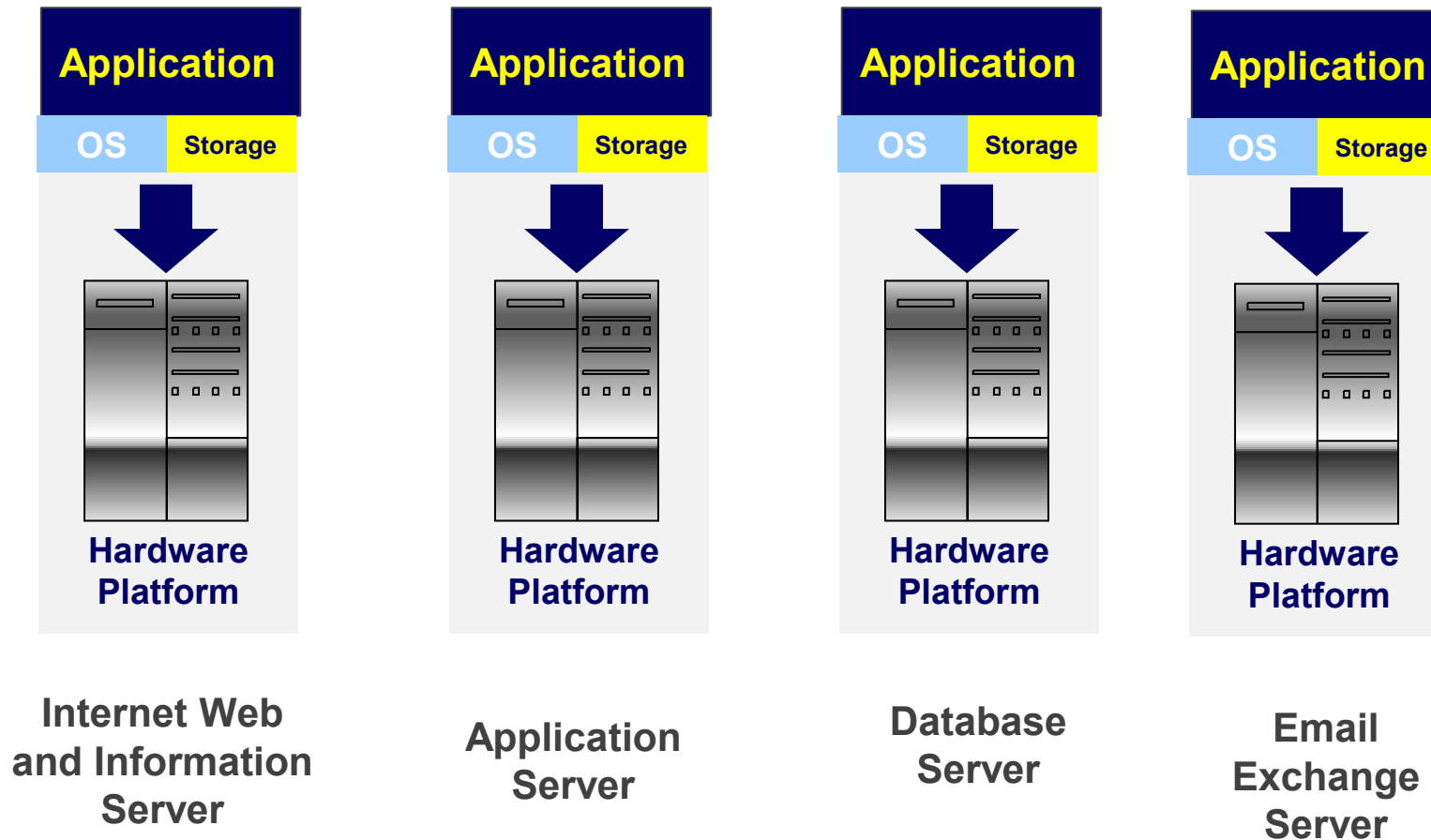
https://www.youtube.com/watch?v=vUUC_eDb2z0



Virtualization In Practice

SERVER CONSOLIDATION

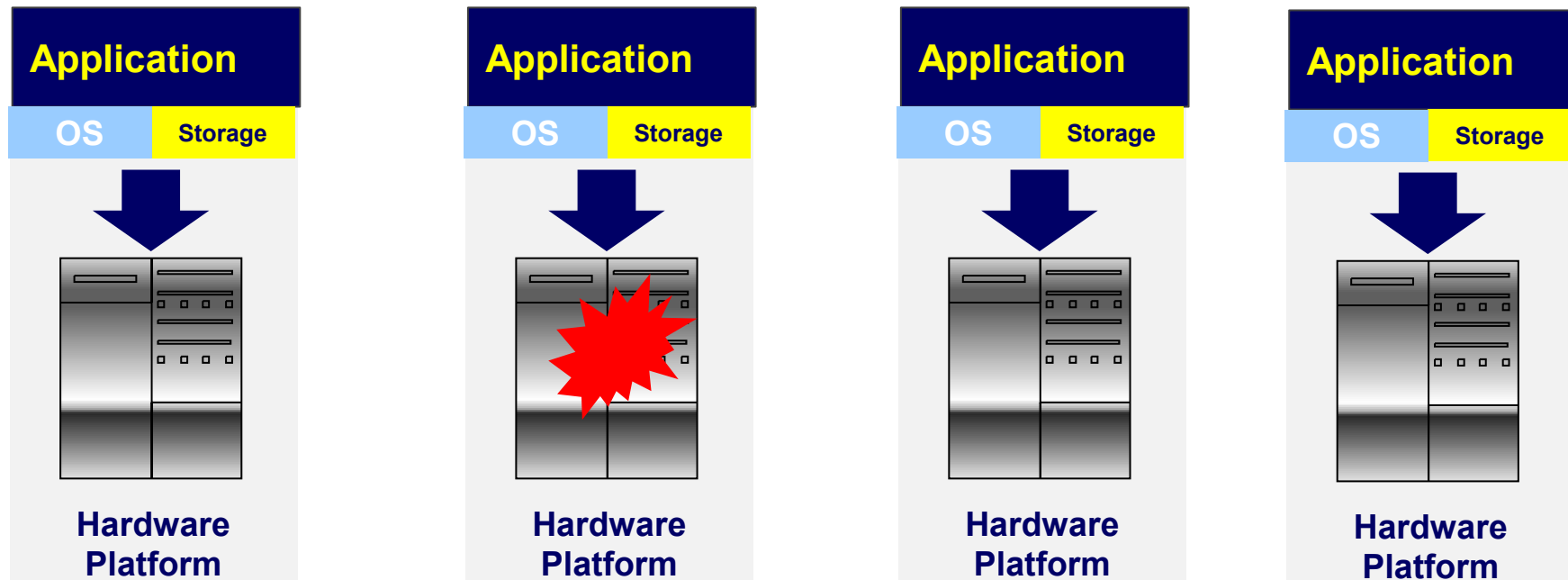
Traditional Server Infrastructure



The Traditional Server Concept

- Servers are viewed as an **integral** computing unit.
 - Each unit includes the hardware, the OS, the storage, and the related applications.
- Servers are often identified and referred to by their **function**.
 - File server, Database server, SQL server, Web server Exchange server, ...
- When current server capacity reaches its limit, **a NEW server** must be added

Server Failure



Internet Web and
Information Server

Application Server

Database Server

Email Exchange
Server

A hardware failure causes service interruption

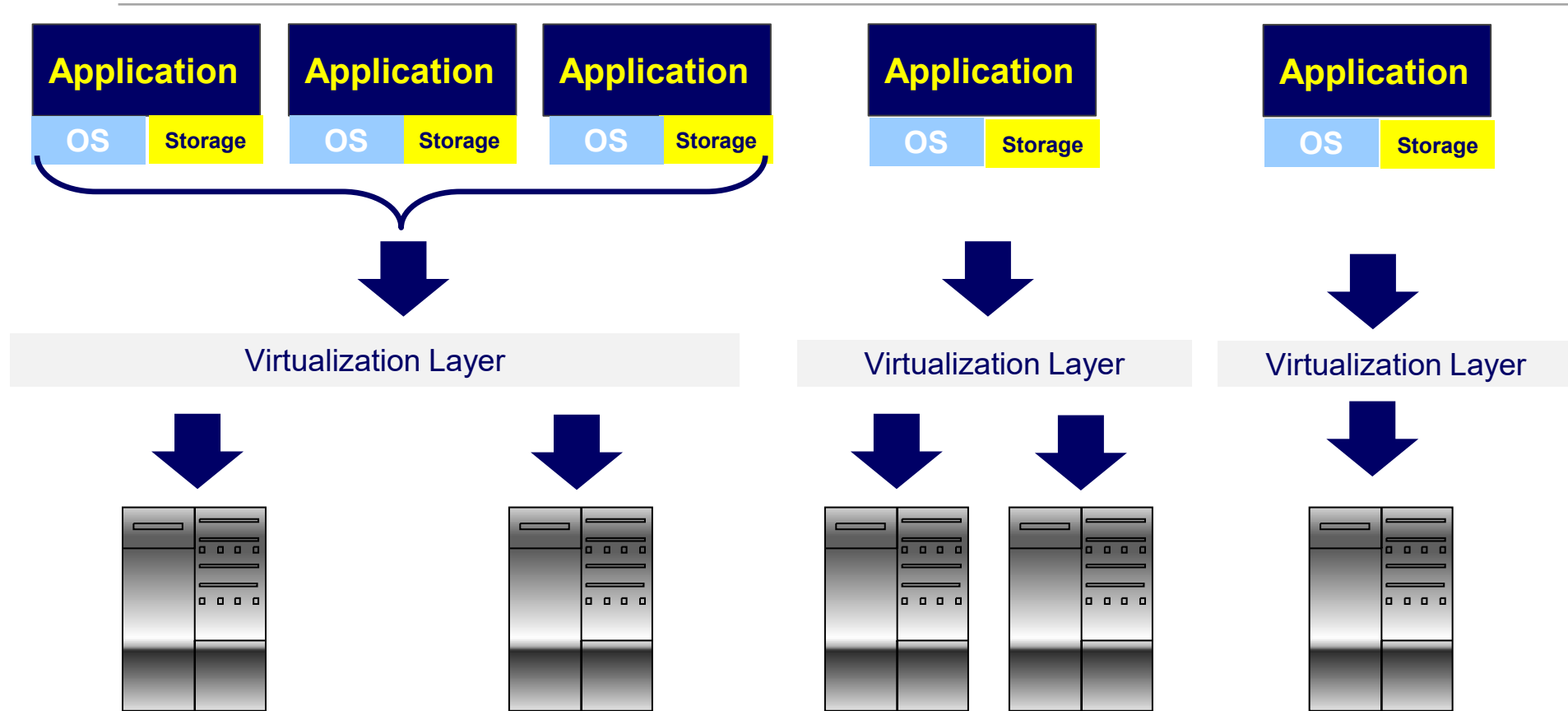


The Traditional Server Concept

Disadvantages

- Maintenance cost is high
 - Acquisition and hardware repair cost
- Replication is challenging
 - Redundancy is costly and difficult to implement
- Scalability may be a limiting factor
- Highly vulnerable to hardware failures
- **Often, utilization is low.**

Virtual Server Infrastructure



Hardware Infrastructure



Server Virtualization

- Server virtualization enable server Consolidation and Containment
 - Eliminating “server sprawl” via deployment of systems as “virtual machines” that can run safely and move transparently across shared hardware
- A virtual server can be serviced by one or more hosts, and one host may house more than one virtual server.
 - This results in increased server utilization rates
 - From 5-15%, traditional servers, to 60-80%



The Virtual Server Concept

- Virtual servers can still be referred to by their **function** i.e., email server, database server, etc.
- If the environment is built correctly, **virtual servers will not be affected by the loss of a host.**
- Hosts may be removed and introduced almost at anytime to accommodate maintenance.



The Virtual Server Concept – Cont'd

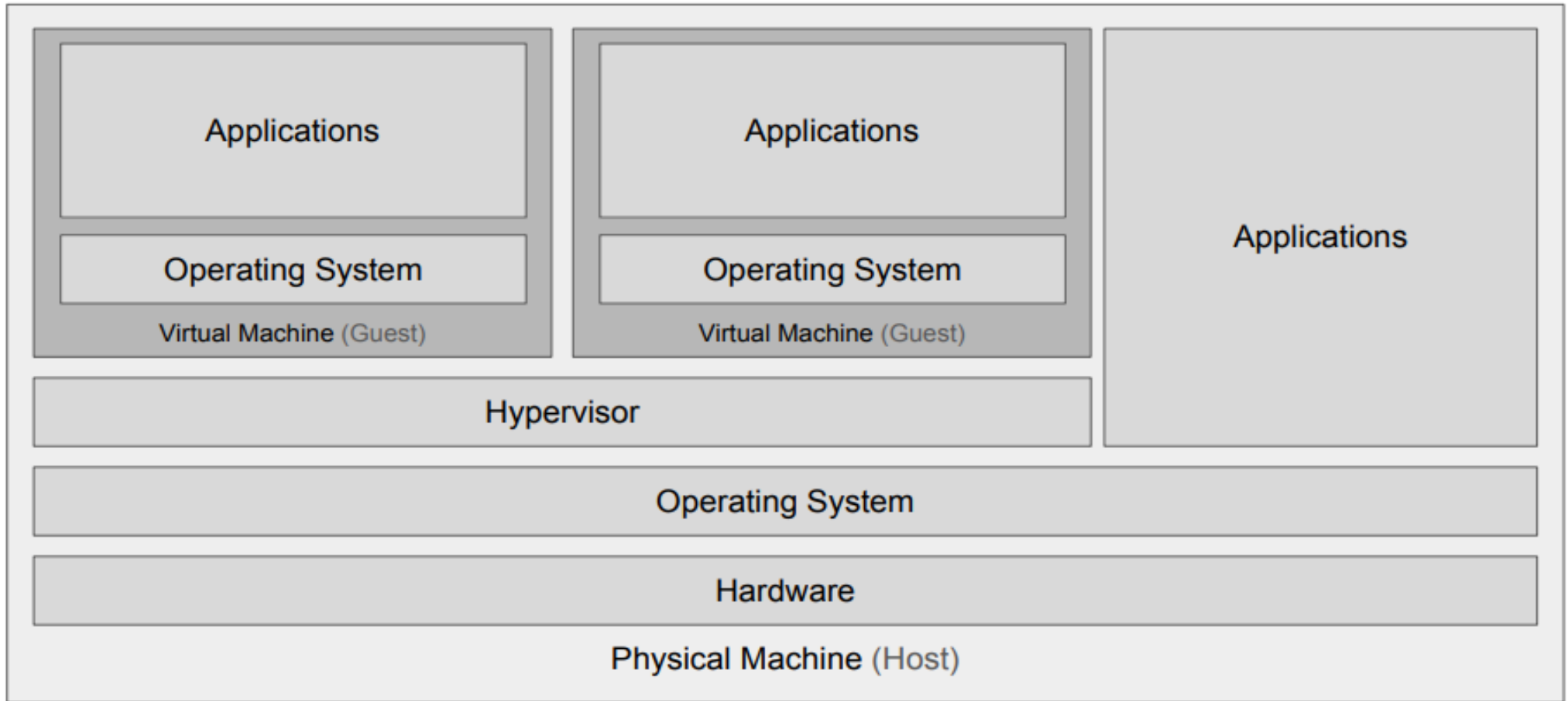
- Virtual servers can be scaled up and down easily.
 - The number of resources allocated to a virtual server can be adjusted dynamically to meet the computation requirements of the virtual server
- Server “**cloning**” can be easily achieved
 - Multiple, identical virtual servers can be easily created based on server templates
- Virtual servers can be migrated from host to host dynamically, as needed.



Virtualization Advantages

- **Resource optimization** that would result in reducing hardware, power and space requirement.
- Virtualization allows for the **quick deployment, migration, and replication** of VMs.
- **Support for Legacy Systems**: Virtualization allows legacy applications to run in a modern cloud environment without requiring significant changes to the underlying infrastructure
- Better **automation**.

Virtualization – How it may look like?!

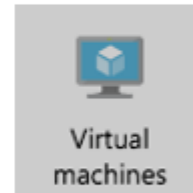




Lab – Use Hosted Hypervisors

VMs in the Cloud

- Getting VMs from
 - AWS EC2
 - Azure
 - Google Cloud





In-class Demo

Create VMs on GCP

Google Cloud Coupons will be provided next week



Notes on Cloud Use

1. You will not be able to run the notebooks on the cloud until you redeem the GCP coupons next week.
2. For now, keep track of the steps to repeat them after you redeem your GCP coupons.
3. Don't use a free trial on GCP for your account. Coupons can't be redeemed on accounts with free trial on them.

Waitlisted Students

- All materials for first two weeks will be uploaded here

